UNDERSTANDING
COMPOUNDS AND REACTIONS

SUBSTANCES CAN CHANGE THEIR FORM IN CHEMICAL REACTIONS

A chemical compound is a substance composed of two or more types of atom held together by chemical bonds. Water, for example, is made of hydrogen atoms bonded to oxygen atoms. Chemical reactions involve the breaking or forming of chemical bonds, resulting in new substances.

POTASSIUM IN WATER
When potassium reacts with water, hydrogen gas is released. The reaction also produces heat, which ignites the hydrogen.

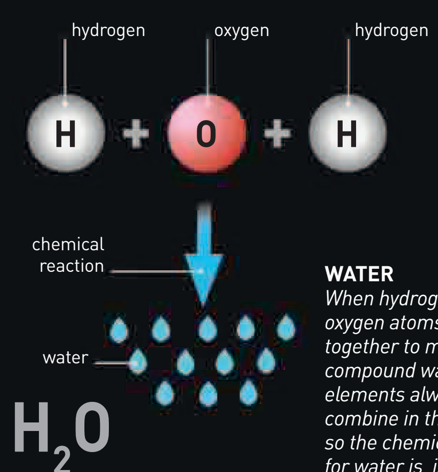
Most solids, liquids, and gases are mixtures of compounds or elements (an element is a substance made of only one kind of atom). Air, for example, is a mixture composed mainly of the elements nitrogen and oxygen, with most of the rest consisting of the element argon and the compounds water, carbon dioxide, and methane. In some compounds, the atoms bond by sharing electrons to form molecules (see right). This kind of bond is called a covalent bond. In other types of compound, the atoms have lost or gained electrons and are in the form of electrically charged ions. These ions are held together by the electrical forces between them: ionic bonds.

1,000

MILLION MILLION MILLION—ROUGHLY THE NUMBER OF **MOLECULES** IN **ONE DROP** OF WATER

COMPOUNDS

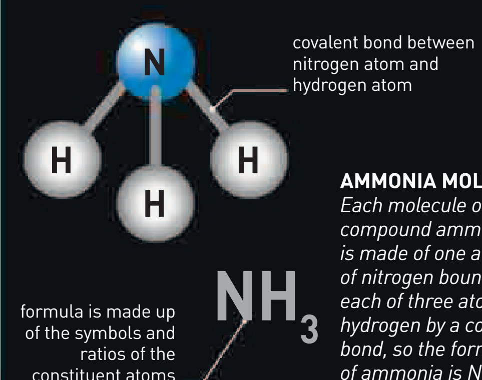
Any sample of a particular compound always has the same ratio of the elements of which it is composed. For example, if the compound methane was broken down into its constituent atoms and the atoms were counted, the carbon (C) and hydrogen (H) atoms would always be in the ratio of 1:4. As a result, every compound has a chemical formula—methane's is CH_4.



WATER
When hydrogen and oxygen atoms react together to make the compound water, the elements always combine in the ratio 2:1, so the chemical symbol for water is is H_2O.

MOLECULES

The atoms that make up a molecule are joined together by one or more covalent bonds, rather than the ionic bonds that hold atoms together in nonmolecular compounds such as sodium chloride (common salt). The smallest molecules are composed of just two atoms, but some are much larger; proteins, for example, may consist of tens of thousands of atoms. Some elements can also exist as molecules. For instance, pure hydrogen and oxygen are typically composed of two-atom (diatomic) molecules—H_2 and O_2.



AMMONIA MOLECULE
Each molecule of the compound ammonia is made of one atom of nitrogen bound to each of three atoms of hydrogen by a covalent bond, so the formula of ammonia is NH_3.

REACTIONS

The elements or compounds that take part in a reaction are called reactants. During a reaction, the bonds of the reactants break and new bonds may form, producing one or more different substances—known as the products. For example, in the reaction shown on the right, the atoms of two reactants combine to form a single compound as the product. The atoms involved in a reaction do not go out of or come into existence—they are just rearranged, so the total mass of the products is the same as the total mass of the reactants.

ENERGETIC REACTION
Two reactants may react spontaneously when mixed, forming a new compound. In some reactions, energy may be released, as when water and potassium react.